MATHEMATICS

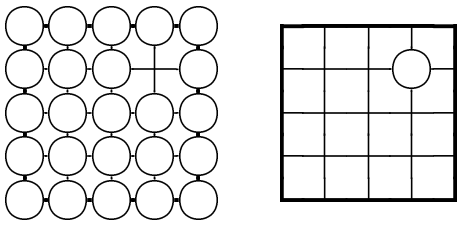
University *of* NORTH ALABAMA

Spring 2025 Math Problem of the Week: Problem #7

Due noon Monday, March 3
Open to all UNA undergraduates

Directions: Deliver your written solution to me (Dr. Bryant) in CPTMTH 328 by noon Monday, March 3 or, if you cannot find me, you may leave it with Ms. Buckley in CPTMTH 377. All work must be your own. You are required to justify your response fully. If you utilize a well-known theorem, state the theorem's full name and proposition, showing that the conditions of the theorem are met. Good luck!

Problem #7: Popularized by the triangular variant found in Cracker Barrels, *peg solitaire* is a board game in which one fills a grid with pegs (traditionally marbles), so that the grid is completely filled but for a single unoccupied position. We call this the *initial board*. The player then proceeds to "jump" a peg orthogonally over an adjacent peg into an available hole two positions away, removing the jumped peg. Play continues until no more jumps are available. A *winning board* is one with a single remaining peg, occupying the initially empty position. A potential set of initial and winning 5x5 boards is shown below.



Show for a game played on a 5x5 grid, that for all possible initial boards, there does not exist a sequence of jumps arriving at a winning board.